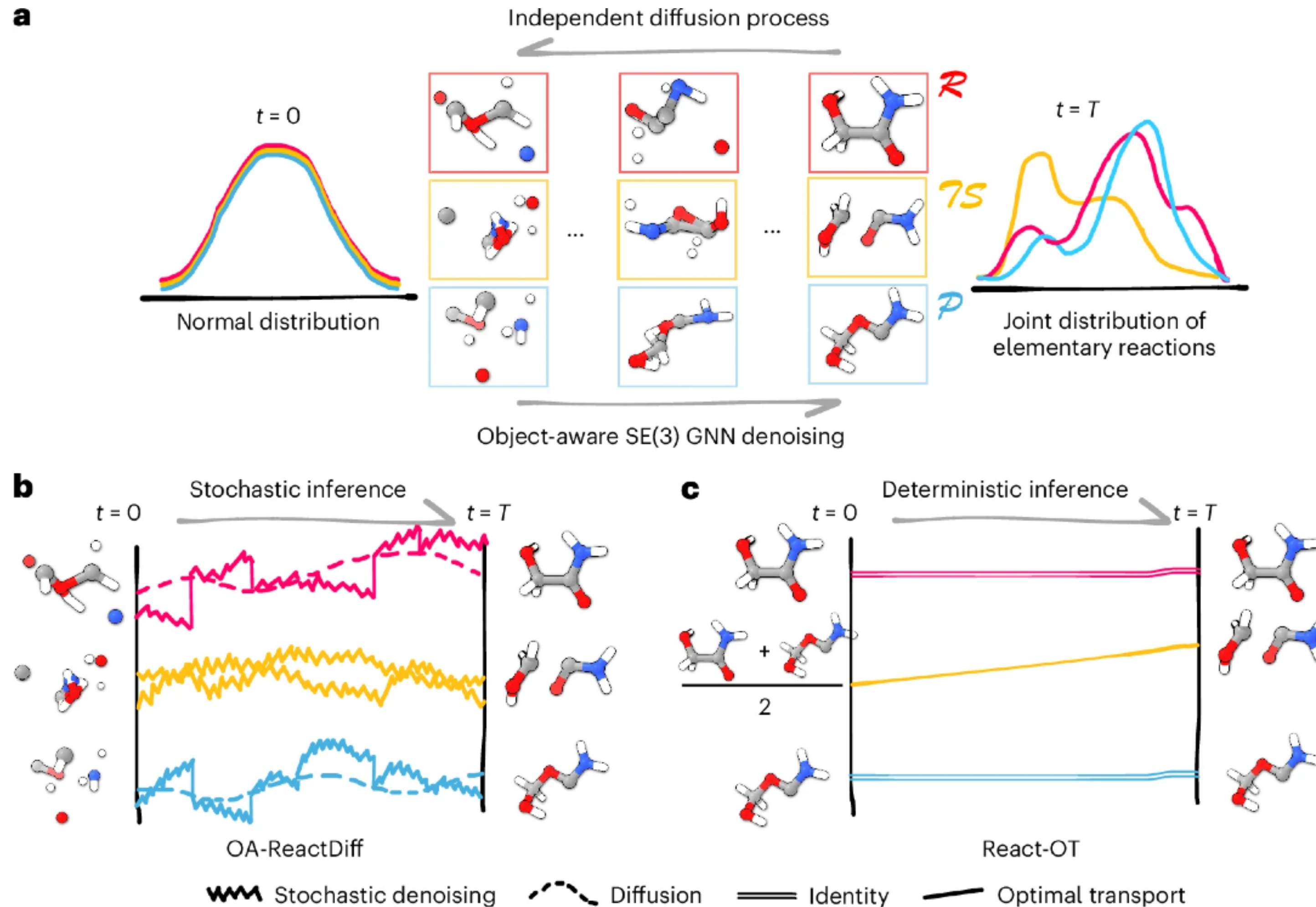


Similar work 1 for generating transition states in chemical reactions [Duan et al. 2025 nature MI]



Here:

- positions modeled as distributions on graphs
- Forward: noising
- SE(3) GNN transition kernel (LEFTNet)

Our work:

- Topological stochastic heat diffusion

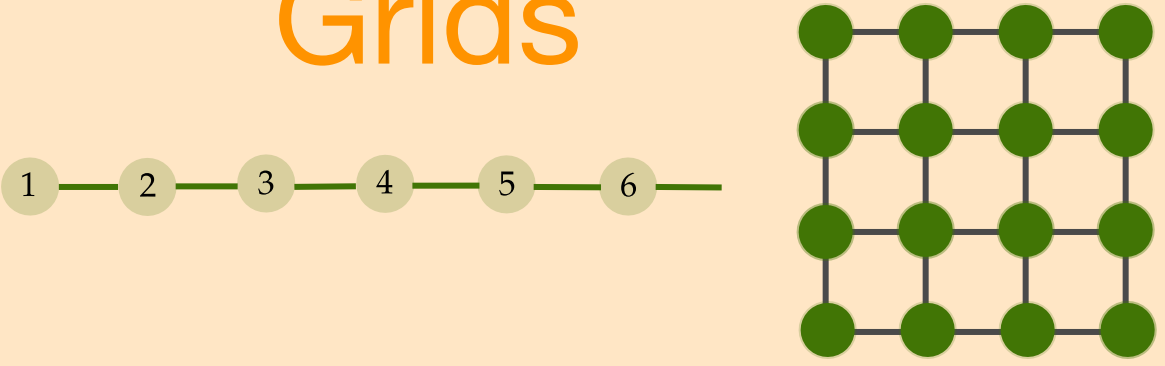
$$dY_t = -cLY_t dt + g_t dW_t$$

- Topology-aware transition kernels

Data, domain, model

Scalar $\mathbb{R}$
Categorical $\mathbb{Z}$
Angle $\mathbb{S}$
high-dim data $\mathbb{R}^n$
3d (coordinates $\mathbb{R}^3$)
 m angles $\mathbb{T}^m$

Grids
1 2 3 4 5 6
Graphs
SCs
Ellipsoid
Sphere
1D time
2D/3D surface



Geometric models: EGNN, GAT, TFN ...

Topological models: GNN, SNN ...